

इंटरनेट

मानक

Disclosure to Promote the Right To Information

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“जानने का अधिकार, जीने का अधिकार”

Mazdoor Kisan Shakti Sangathan

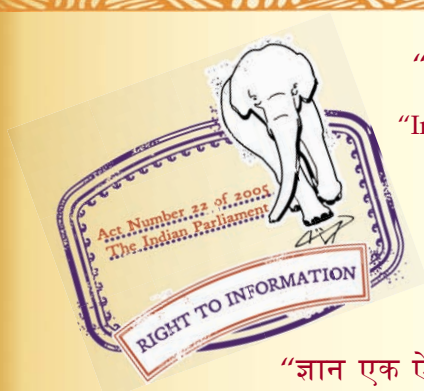
“The Right to Information, The Right to Live”

“पुराने को छोड़ नये के तरफ”

Jawaharlal Nehru

“Step Out From the Old to the New”

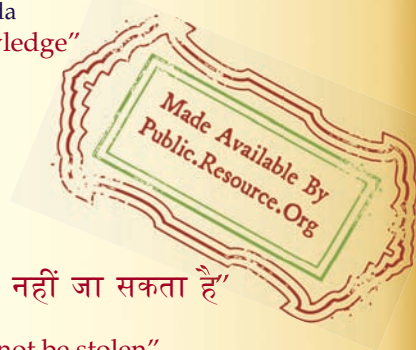
IS 2112 (2003): Silver and Silver Alloys,
Jewellery/Artefacts - Fineness and Marking [MTD 10:
Precious Metals]



“ज्ञान से एक नये भारत का निर्माण”

Satyanarayan Gangaram Pitroda

“Invent a New India Using Knowledge”



“ज्ञान एक ऐसा खजाना है जो कभी चुराया नहीं जा सकता है”

Bhartrhari—Nitiśatakam

“Knowledge is such a treasure which cannot be stolen”

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भारतीय मानक
चाँदी एवं चाँदी मिश्रधातुएँ, आभूषण/शिल्पकारी —
शुद्धता एवं मुहरांकन — विशिष्टि
(दूसरा पुनरीक्षण)

Indian Standard
SILVER AND SILVER ALLOYS, JEWELLERY/
ARTEFACTS — FINENESS AND MARKING —
SPECIFICATION
(*Second Revision*)

ICS 39.060

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI 110002

FOREWORD

This Indian Standard (Second Revision) was adopted by the Bureau of Indian Standards, after the draft finalized by the Precious Metals Sectional Committee had been approved by the Metallurgical Engineering Division Council.

This standard was published earlier in 1962 and later revised in 1981. In this revision, apart from the existing fine grades 9999, 9995, 999, 970, 925 three new grades 900, 835, 800 internationally in vogue have been added keeping in mind the requirements of international trade. The standard has therefore been harmonized with ISO 9202-1991 'Jewellery — Fineness of precious metals alloys'.

In many countries, the silver jewellery/artefacts (coins, medals, utensils, etc) traded are certified for purity and are marked with various details like purity/fineness of the article, manufacturer's mark, the mark of the assaying laboratory responsible for certifying the purity, etc. These jewellery/artefacts are termed as Hallmarked.

The need for availability of jewellery/artefacts of silver of certified purity in the country, for domestic use and for export is the major reason for undertaking the preparation of this standard.

This standard is primarily meant to classify the various grades of silver according to their silver content and to provide guideline to mark the silver jewellery/artefacts.

The composition of the Committee responsible for the formulation of this standard is given in Annex A.

For the purpose of deciding whether a particular requirement of this standard is complied with the final value, observed or calculated, expressing the result of a test or analysis, shall be rounded off in accordance with IS 2 : 1960 'Rules for rounding off numerical values (*revised*)'. The number of significant places retained in the rounded off value should be the same as that of the specified value in this standard.

**AMENDMENT NO. 4 JULY 2008
TO
IS 2112 : 2003 SILVER AND SILVER ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS
AND MARKING — SPECIFICATION**

(Second Revision)

(Page 1, clause 5) — Insert the new Grade 990 in between 999 and 970 as follows:

<i>Grade</i>		<i>Fineness</i>
990	Silver alloys for Jewellery artefacts	990.0

(Page 2, clause 7.1, line 2) — Substitute the word ‘this case’ for ‘the case’.

(Page 2, clause 7.1, line 5) — Substitute ‘The hallmarked articles shall carry the following markings:’ for ‘The mark shall comprise of the following:’.

(Page 2, clause 7.1) — Substitute the following for the existing clause 7.1(a) to 7.1(e):

- a) BIS mark,
- b) Purity grade/fineness,
- c) Assay Centre’s Identification mark,
- d) Year of marking denoted by a letter symbol (as defined by BIS);
and
- e) Jeweller’s mark/manufacturer’s identification mark.

(Page 2, clause 7.2, Note, line 1) — Substitute ‘identification mark’ for ‘logo’.

[Page 2, clause 7.3(c) line 5] — Add the following in continuation after the word ‘strength’:

‘unless the weight of silver is declared by the jeweller and marked on the article by the centre’.

(MTD 10)

AMENDMENT NO. 3 FEBRUARY 2006
TO
IS 2112 : 2003 SILVER AND SILVER ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION

(Second Revision)

(Page 2, clause 6.3.1, line 1) — Substitute the word 'detection' for 'Determination'.

(MTD 10)

Reprography Unit, BIS, New Delhi, India

**AMENDMENT NO. 1 DECEMBER 2003
TO
IS 2112 : 2003 SILVER AND SILVER ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION**

(*Second Revision*)

(*Page 1, clause 5, Note*) — Insert 'and lead' after 'cadmium' at the end.

(MTD 10)

Reprography Unit, BIS, New Delhi, India

**AMENDMENT NO. 2 JANUARY 2004
TO
IS 2112 : 2003 SILVER AND SILVER ALLOYS,
JEWELLERY/ARTEFACTS — FINENESS AND
MARKING — SPECIFICATION**

(Second Revision)

(Page 1, clause 1.2, Note) — Substitute the following for the existing note:

NOTE — Artefacts cover medallions, utensils and coins provided the purity of silver alloy for artefacts is as per 5.

(Page 2, clause 6.3) — Insert the following new clause after 6.3:

'6.3.1 Determination of cadmium and lead in silver and silver alloys shall be carried out by X-ray fluorescence (XRF) method till an established test procedure is formulated.'

(MTD 10)

Reprography Unit, BIS, New Delhi, India

Indian Standard

SILVER AND SILVER ALLOYS, JEWELLERY/ ARTEFACTS — FINENESS AND MARKING — SPECIFICATION

(*Second Revision*)

1 SCOPE

1.1 This standard specifies:

- a) Three grades of fine silver, and
- b) Five grades of silver alloys used in the manufacture of jewellery/artefacts of silver based on their silver content.

1.2 This standard also specifies the guidelines for marking of purity and other details on tested silver jewellery/artefacts.

NOTE — Artefacts covers medallions and coins provided the purity of silver alloy for artefacts is as per 5.

2 REFERENCES

The following standards contain provisions which through reference in this text, constitute provisions of this standard. At the time of publication, the editions indicated were valid. All standards are subject to revision and parties to agreement based on this standard are encouraged to investigate the possibility of applying the most recent editions of the standards indicated below.

<i>IS No.</i>	<i>Title</i>
1387 : 1993	General requirements for the supply of metallurgical materials (<i>second revision</i>)
2113 : 2002	Assaying silver in silver and silver alloys — Methods (<i>second revision</i>)

3 TERMINOLOGY

For the purpose of this standard, the following definitions shall apply.

3.1 Assaying — The method of accurate determination of silver content of the sample expressed in parts per thousand (‰).

3.2 Fineness — The ratio between the mass of silver content and total mass expressed in parts per thousand (‰).

3.3 Silver — The metallic element silver, free from any other element.

3.4 Silver Alloy — Silver alloyed with one or more other elements (*see Note in 5*).

4 SUPPLY OF MATERIAL

General requirements relating to the supply of material are laid down in IS 1387.

5 GRADES

Fine silver and silver alloys shall be classified in accordance with the following grades depending upon their fineness:

<i>Grade</i>		<i>Fineness</i>
9999	} Fine silver	999.9
9995		999.5
999		999.0
970	} Silver alloys for jewellery artefacts	970.0
925		925.0
900		900.0
835		835.0
800		800.0

NOTE — The above classification of silver alloys is applicable for silver jewellery/artefacts also. All the above grades of silver and silver alloys shall be free from cadmium.

6 REQUIREMENTS

6.1 Silver alloys used for manufacture of jewellery/artefacts shall have the fineness as given in 5 without any negative tolerance.

6.1.1 Solder

Solder used for manufacture of jewellery/artefact shall be of purity 750 silver minimum for grades 900, 925 and 970 and 650 silver minimum for grades 800 and 835 respectively and shall be free from cadmium. However, melting purity of articles shall not be less than the purity for which it has been offered for hallmarking as per the grades given in 5.

6.2 Sampling

Unless otherwise specified for the jewellery/artefacts of the same design and quality, the sample shall be taken from 10 percent of the semi-finished/finished jewellery/artefacts selected at random. In all other cases, the sample shall be taken from each jewellery/artefacts. Any other sampling plan may also be followed subject to the agreement between the contracting parties or as specified by the inspection agency as the case may be.

6.3 Assaying

The sample taken as prescribed in 6.2 shall be assayed in accordance with IS 2113. The actual weight of samples, etc shall, however, be as prescribed in respective test methods as detailed in IS 2113.

6.4 Tolerances

No negative tolerance shall be allowed on the silver content of silver jewellery/artefact sampled and tested as per 6.2 and 6.3 respectively.

7 MARKING

7.1 The silver and silver alloys, jewellery/artefacts shall be stamped with the Standard Mark in the case known as the 'HALLMARK' by BIS recognized Assaying and Hallmarking Centre only. The mark shall comprise of the following:

- a) BIS logo;
- b) Purity grade/fineness;
- c) Assay centre's logo/symbol;
- d) Year of marking denoted by a letter symbol (as defined by BIS); and
- e) Jeweller's mark/manufacturer's identification mark.

7.2 Hallmarking

The use of Hallmark is governed by the provisions of the *Bureau of Indian Standards Act, 1986* and Rules

and Regulations made thereunder. The details of conditions under which the licence for the use of the Hallmark may be granted to jewellers or jewellery/artefact manufacturers may be obtained from the Bureau of Indian Standards.

NOTE — Jeweller's/Sponsor's logo on each article shall be marked before offering the lot to BIS recognized assaying and Hallmarking Centre for Hallmarking.

7.2.1 Only linear designs shall be permitted.

7.2.2 Markings shall be legible and durable. Marking shall be visible by a magnifying glass of magnification 10 X.

7.2.3 Markings shall be done on all parts which can be easily removed or replaced.

7.3 Silver jewellery/artefacts on which marking should not be applied are as follows:

- a) Any silver jewellery/artefact which has not been assayed by BIS recognized Assaying and Hallmarking Centre.
- b) Any silver jewellery/artefact which fails to comply with specified fineness.
- c) Any silver jewellery/artefact which is not solid or which is made with hollow centre and then filled with base metal, cement, lac or other substance thereby giving the article a false or added weight or strength.
- d) Silver jewellery/artefact on which it is physically not possible to stamp the marks.

ANNEX A**(Foreword)****COMMITTEE COMPOSITION****Precious Metals Sectional Committee, MTD 10**

<i>Organization</i>	<i>Representative(s)</i>
In personal capacity (<i>Sopan Bang, Co-operative Housing Society, Flat No. 9, Building No. A-1-A, Pashan, Panchabati Pune 411008</i>)	SHRI S. C. SAHA (<i>Chairman</i>)
All India Handicrafts Board, New Delhi	SHRI S. K. MOHAPATRA SHRI R. P. CHUGH (<i>Alternate</i>)
All India Sarafa Association, Delhi	SHRI PREM NATH KHANNA
Bhabha Atomic Research Centre, Mumbai	DR T. P. RADHAKRISHNAN DR R. G. DHANANESHWAR (<i>Alternate</i>)
Consumer Guidance Society of India, Mumbai	SHRI N. G. WAGLE DR A. R. CHENOY (<i>Alternate</i>)
Consumer Education and Research Centre, Ahmedabad	DR C. J. SHISHOO
Gujarat Gold Centre, Ahmedabad	SHRI SURESH C. KANSARA SHRI ANIL C. KANSARA (<i>Alternate</i>)
Gujarat Bullion Refinery Ltd, Ahmedabad	SHRI J. N. KANSARA SHRI V. J. KANSARA (<i>Alternate</i>)
Hindustan Platinum Pvt Ltd, Mumbai	SHRI CHANDRAKANT S. CHOKSI SHRI SHASHIKANT S. CHOKSI (<i>Alternate</i>)
Hutti Gold Mines Co Ltd, Karnataka	SHRI J. S. GURURAJA NAIK DR M. L. PATIL (<i>Alternate</i>)
Jetha Bhai Jhaveri, New Delhi	SHRI VIMAL JETHA
Medical Education Research (Dental), Mumbai	DR V. M. DESAI
MMTC Ltd, New Delhi	SHRI D. K. MANCHANDA SHRI SANJAY ANAND (<i>Alternate</i>)
National Refinery Pvt Ltd, Mumbai	SHRI CHANDRAKANT K. SHAH SHRI S. P. SONAWALAQA (<i>Alternate</i>)
National Test House, Kolkata	SHRI A. BISWAS SHRIMATI SHASHI SHARMA (<i>Alternate</i>)
Surabhi, Hyderabad	SHRI S. SYAMSUNDER
Tamilnadu Jeweller's Federation, Chennai	DR VASANTIAL C. MEHTA SHRI R. CHANDRASEKARAN (<i>Alternate</i>)
The Bombay Bullion Association Ltd, Mumbai	SHRI AMRITLAL G. SONAWALA SHRI SHANTILAL NARANDAS (<i>Alternate</i>)
The Gem and Jewellery Export Promotion Council, Mumbai	SHRI NAND KISHORE M. CHHABRIA SHRI DILIP SHAH (<i>Alternate</i>)
Titan Industries Ltd, Tamil Nadu	SHRI M. KUMAR
World Gold Council, New Delhi	SHRI G. P. PILLAI
BIS Directorate General	SHRI JAGMOHAN SINGH, Director & Head (MTD) [Representing Director General (<i>Ex-officio</i>)]

Member Secretary

SHRI B. K. MUKHOPADHYAY
Joint Director (MTD), BIS

Bureau of Indian Standards

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Amendments are issued to standards as the need arises on the basis of comments. Standards are also reviewed periodically; a standard along with amendments is reaffirmed when such review indicates that no changes are needed; if the review indicates that changes are needed, it is taken up for revision. Users of Indian Standards should ascertain that they are in possession of the latest amendments or edition by referring to the latest issue of 'BIS Catalogue' and 'Standards: Monthly Additions'.

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Amendments Issued Since Publication

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